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1. First run with Demo

The fastest way to verify the install is to use the bundled simulator — no real hardware, no extra terminals. Launch the diagnostic, then:

1. **Help** → **Start demo...**
2. Pick a slave count (default 3, range 1–32).
3. The diagnostic spawns that many `vrmc_sim` processes on UDP multicast `239.192.0.42:23400` with sequential node IDs (5, 6, 7, ...) and **auto-connects**.
4. The slave list populates, MotorView lights up for the first slave, and the telemetry pane starts plotting position / velocity / torque.

The simulator runs entirely inside the diagnostic's own build tree (`vrmc_sim` next to `vr_mc_diagnostic`), so this workflow works straight out of the box — no need to compile or run anything from `vr-mc-sdk` separately.

To stop:

- **Help** → **Stop demo** — terminates the sim processes cleanly.
- Pressing the toolbar **Disconnect** also stops the demo (Help menu state flips back to "Start demo" enabled).
- Closing the diagnostic kills any surviving sim child processes so nothing is left listening on the multicast group.

Trying the Control tab against the demo

Once auto-connect lands, the first slave is selected and you can drive it straight away:

1. Click **Bringup** in the Control tab — the state readout strip walks `SWITCH_ON_DISABLED` → `READY_TO_SWITCH_ON` → `SWITCHED_ON` → `OPERATION_ENABLED` (badge turns green).
2. Type a value into the Setpoint spinbox (default unit = rad in Position mode) and click **Send** — the MotorView needle swings to the new angle. Try the preset buttons `+π/4` / `-π/4` / `Home` for one-click setpoints.
3. Tick `[deg]` in the readout strip to switch the spinbox + slider
4. tracking-error display to degrees.
5. Move the slider with **stream live** checked to scrub the setpoint in real time.
6. **E-STOP** (toolbar, far right) — instantly disarms every slave and pops the panel buttons back to disabled.

If anything in this sequence fails, look at the Log dock at the bottom: every SDO error, PDO online event, and demo-process lifecycle line lands there with a timestamp.

2. Connecting to drives

Connection → **Connect...** (`Ctrl+K`).

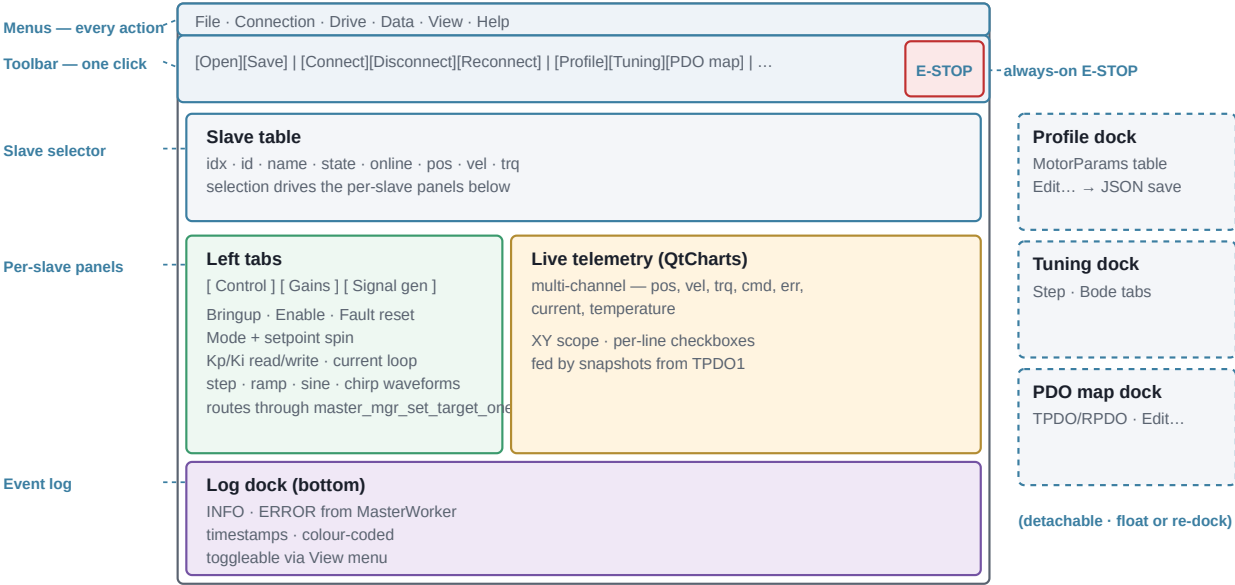
Transport	Fields	When
UDP multicast	Group, Port	Loopback against sim, multi-host bench
ZLG USB-CANFD	Channel, bitrate, FD-bitrate	Real drives

Both take **First node id + Count**. The diagnostic registers that range sequentially into `master_mgr`.

Reconnect (`Ctrl+R`) replays the last config. **Disconnect** is idempotent — worker thread survives for the next connect.

3. Main window tour

Main window — what lives where



File
Connection
Drive
Data
View
Help

Open Profile...
Save Profile

Connect...
Disconnect
Reconnect

Profile
Tuning
PDO map

Configure Drive...
Upload Firmware...
Record telemet

	idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000	0.0000

Control
Gains
Signal gen

(select a slave above)

Bringup
Enable
Disable
Fault reset

Mode

NONE
Apply

Setpoint

Position (rad)
0.0000
Send

☒ pos (rad)
☐ cmd trq

☒ vel (rad/s)
☐ err

☒ trq (Nm)
☐ current (A)

☐ cmd
☐ temp

Live telemetry

pos (rad)
vel (rad/s)
trq (Nm)

Log

```

12:56:05.684 INFO connected: 1 slave(s)
12:56:05.704 INFO PDO online: slave 0 - rx=66, statusword=0x0250

```

Toolbar groups (L → R): File · Connection · View (dock toggles) · Drive · Data · Help · E-STOP (always right-anchored, `Space` shortcut).

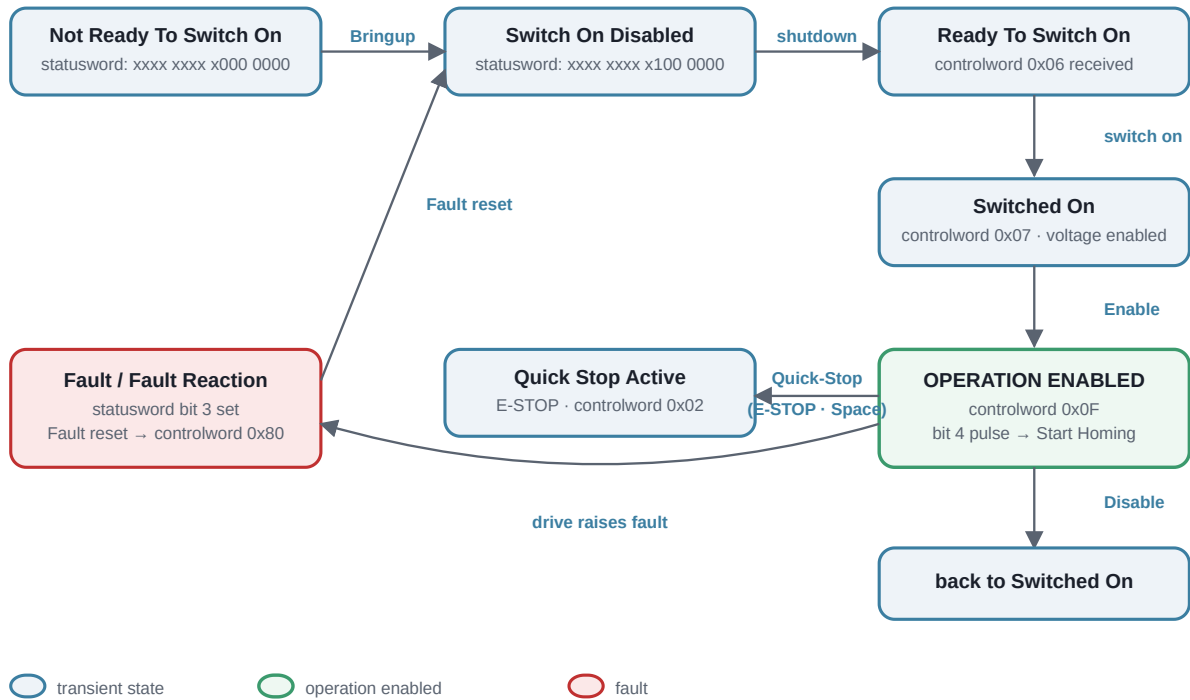
Slave table = one row per slave. Selection drives every per-slave panel.

4. Left-tab panels

Control

Drives the CiA 402 state machine:

CiA 402 drive state machine — what Control-tab buttons do



- **Bringup** — Not Ready → Switched On
- **Enable** — → Operation Enabled (only motion-accepting state)
- **Disable** — back to Switched On (torque off)
- **Fault reset** — pulses CW bit 7 → Switch On Disabled

Mode selector + **Setpoint** spin box route through `master_mgr_set_target_one`.

Gains

File
Connection
Drive
Data
View
Help

Open Profile...
Save Profile
Connect...
Disconnect
Reconnect
Profile
Tuning
PDO map
Configure Drive...
Upload Firmware...
Record telemet

	idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000	0.0000

Control
Gains
Signal gen

Loop
Kp
Ki

Current
0.000000
0.000000
Read
Apply

Velocity
0.000000
0.000000
Read
Apply

Position
0.000000
0.000000
Read
Apply

Read all

☒ pos (rad)
☐ cmd trq

☒ vel (rad/s)
☐ err

☒ trq (Nm)
☐ current (A)

☐ cmd
☐ temp

Live telemetry

pos (rad)
vel (rad/s)
trq (Nm)

Log

Per-loop (current / velocity / position) Kp+Ki. Read pulls current values via SDO; Write pushes back.

Signal generator

FileConnectionDriveDataViewHelp

Open Profile...

Save Profile

Connect...

Disconnect

Reconnect

Profile

Tuning

PDO map

Configure Drive...

Upload Firmware...

Record telemet

	idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000	0.0000

ControlGainsSignal gen

(select a slave above)

Waveform

Step

Target

Position (rad)

Amplitude

1.0000

Offset

0.0000

Frequency

1.000 Hz

Frequency end

5.000 Hz

Duration (0=∞)

5.00 s

Arm / Run

Stop

☒ pos (rad)

☐ cmd trq

☒ vel (rad/s)

☐ err

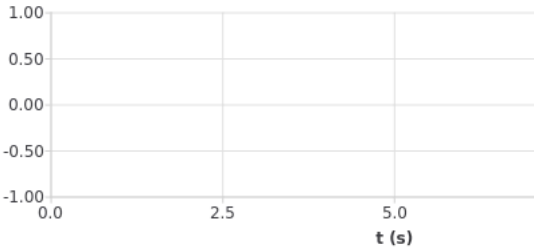
☒ trq (Nm)

☐ current (A)

☐ cmd

☐ temp

Live telemetry



pos (rad)

vel (rad/s)

trq (Nm)

Log

12:56:08.530 INFO connected: 1 slave(s)
12:56:08.550 INFO PDO online: slave 0 - rx=66, statusword=0x0250

Waveforms: constant · step · ramp · sine · chirp. Chirp feeds the Bode sweep.

5. Detachable panes

Created hidden + floating. Toolbar View group brings them out.

Profile

FileConnectionDriveDataViewHelp

Open Profile...

Save Profile

Connect...

Disconnect

Reconnect

Profile

Tuning

PDO map

Configure Drive...

Upload Firmware...

Record telemet

	idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000	0.0000

ControlGainsSignal gen

(select a slave above)

Bringup

Enable

Disable

Fault reset

Mode

NONE

Apply

Setpoint

Position (rad)

0.0000

Send

☒ pos (rad)

☒ vel (rad/s)

☒ trq (Nm)

☐ cmd pos

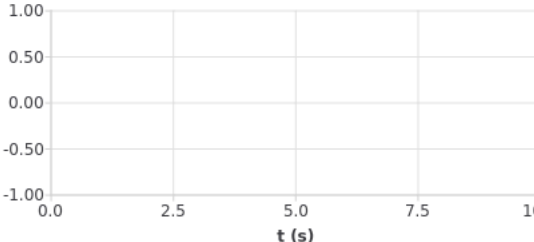
☐ cmd trq

☐ err

☐ current (A)

☐ temp (°C)

Live telemetry



pos (rad)vel (rad/s)trq (Nm)

Log

12:56:17.539 INFO connected: 1 slave(s)
12:56:17.559 INFO PDO online: slave 0 - rx=66, statusword=0x0250

Read-only 15-row `MotorParams` table (type, pole pairs, Rs, Ls_d, Ls_q, rated flux, inertia, voltage/current limits). **Edit...** opens a modal; File → Save Profile persists to JSON.

Tuning

File
Connection
Drive
Data
View
Help

Open Profile...
Save Profile
Connect...
Disconnect
Reconnect
Profile
Tuning
PDO map
Configure Drive...
Upload Firmware...
Record telemet

	idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000	0.0000

Control
Gains
Signal gen

(select a slave above)

Bringup
Enable
Disable
Fault reset

Mode

NONE
Apply

Setpoint

Position (rad)
0.0000
Send

☒ pos (rad)
☒ vel (rad/s)
☒ trq (Nm)
☐ cmd pos
☐ cr

☐ cmd trq
☐ err
☐ current (A)
☐ temp (°C)

Live telemetry

1.00
0.50
0.00
-0.50
-1.00

0.0 2.5 5.0 7.5 10

t (s)

pos (rad) vel (rad/s) trq (Nm)

Log

12:56:20.534 INFO connected: 1 slave(s)

12:56:20.553 INFO PDO online: slave 0 - rx=62, statusword=0x0250

- **Step** — target, step size, offset, duration → run → overshoot, rise time, SS error
- **Bode** — log-sweep chirp + Goertzel correlation → magnitude/phase vs. frequency

PDO map

FileConnectionDriveDataViewHelp

Open Profile...

Save Profile

Connect...

Disconnect

Reconnect

Profile

Tuning

PDO map

Configure Drive...

Upload Firmware...

Record telemet

	idx	id	name	state	online	pos (rad)	vel (rad/s)	trq (Nm)
0	0	5	cia402-node-5	disabled	yes	0.0000	0.0000	0.0000

ControlGainsSignal gen

(select a slave above)

BringupEnableDisableFault reset

Mode

NONEApply

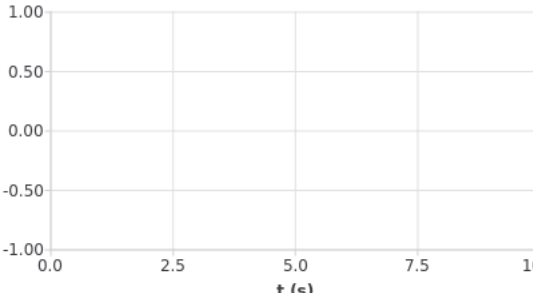
Setpoint

Position (rad)0.0000Send

☒ pos (rad)☒ vel (rad/s)☒ trq (Nm)☐ cmd pos☐ cr

☐ cmd trq☐ err☐ current (A)☐ temp (°C)

Live telemetry



pos (rad)vel (rad/s)trq (Nm)

Log

12:56:14.533 INFO connected: 1 slave(s)
12:56:14.552 INFO PDO online: slave 0 - rx=62, statusword=0x0250

Two sections (TPDO1 / RPDO1). Offset, OD index, sub, bits, name, live value. **Edit...** runs the CiA 301 S7.2.2.2 remap dance via SDO.

6. Configure Drive

Drive → **Configure Drive...** (Ctrl+Shift+C).

Slave #0 — cia402-node-5

Homing

Motion profile

Protection

Encoder

Method (0x6098)

0 - no homing

Home offset (0x607C)

0 counts

Switch search speed (0x6099:1)

100 cnt/s

Zero search speed (0x6099:2)

10 cnt/s

Homing acceleration (0x609A)

1000 cnt/s²

Start Homing

Zero encoder here

Homing: —

Read from drive

Apply

OK

Cancel

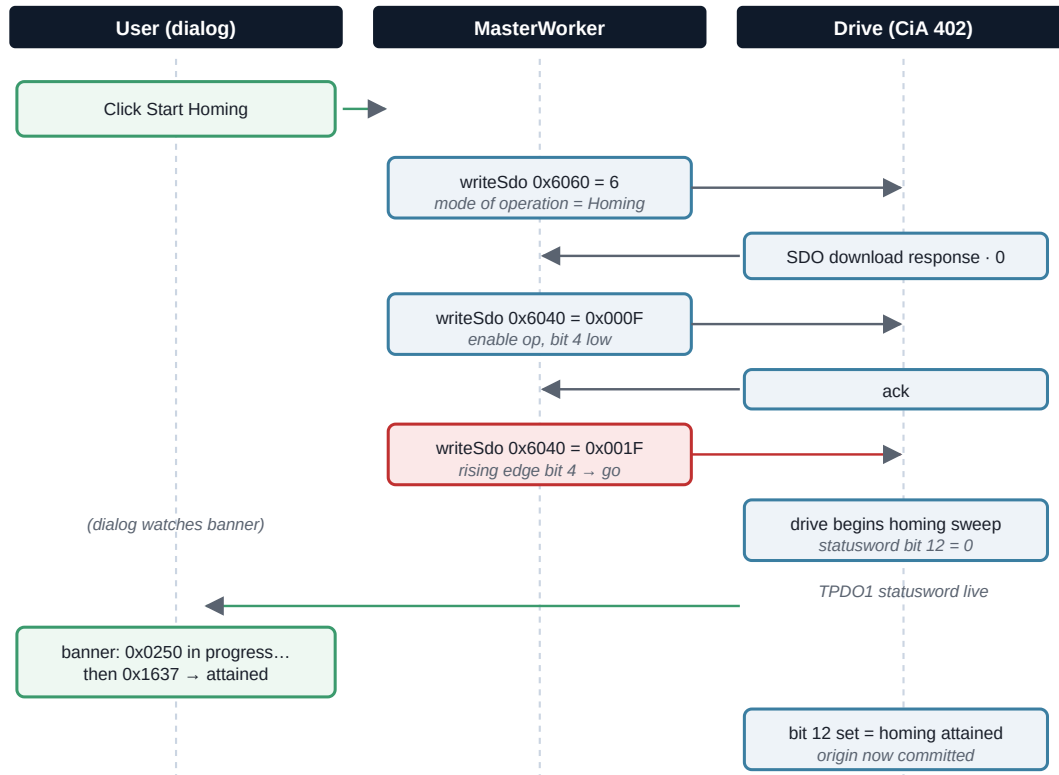
Modeless tabbed dialog. Opens with the selected slave, kicks off a batch SDO read so the form reflects the drive.

6.1 Homing

Field	OD
Method	0x6098
Home offset	0x607C
Switch search speed	0x6099:1
Zero search speed	0x6099:2
Homing acceleration	0x609A

Start Homing — mode=6, controlword 0x000F → 0x001F :

Start Homing — SDO writes + live statusword readback



Zero encoder here — reads `0x6064`, writes `0x607C`, updates form.

6.2 Motion profile

Profile velocity `0x6081`, accel `0x6083`, decel `0x6084`, quickstop decel `0x6085`.

6.3 Protection

Following error `0x6065` + timeout `0x6066`, pos limits `0x607D:1,2`, max speed `0x6080`, max torque `0x6072`. **Zero torque here** reads `0x6077`, writes `0x60B2`.

6.4 Encoder

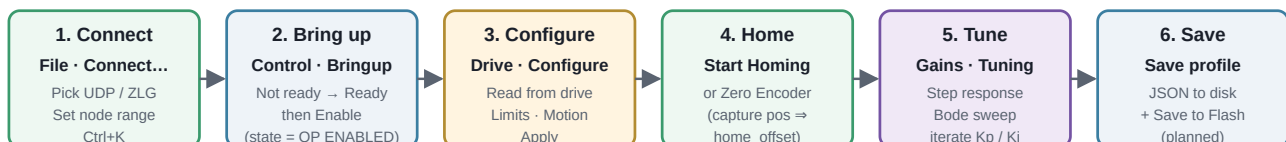
Resolution `0x608F:1`, gear ratio `0x6091:1,2`, feed constant `0x6092:1,2`.

Buttons

- **Read from drive** — batch SDO upload; missing fields reported, not aborted
- **Apply** — batch SDO download
- **OK** — Apply + close · **Cancel** — discard

7. Commissioning workflow

Commissioning workflow — drive out of the box to tuned



At any point: Space (E-STOP) drives every slave to Quick-Stop Active.
Log dock shows info / error · Telemetry chart ticks at the TPDO rate throughout.

1. **Connect** — Ctrl+K → transport → node range.
2. **Bring up** — Control tab → Bringup → Enable.

3. **Configure** — Drive → Configure → Read → set limits → Apply.
4. **Home** — Start Homing (method-driven) or Zero encoder here.
5. **Tune** — Gains Read → Tuning step/Bode → iterate Kp/Ki.
6. **Save** — File → Save Profile + Drive → Save to Flash (planned).

Safety. Keep `Space` reachable the first time you enable a new drive. Wrong limits + wrong mode = motion outside the safe envelope in one control cycle.

8. Tuning workflows

Step response

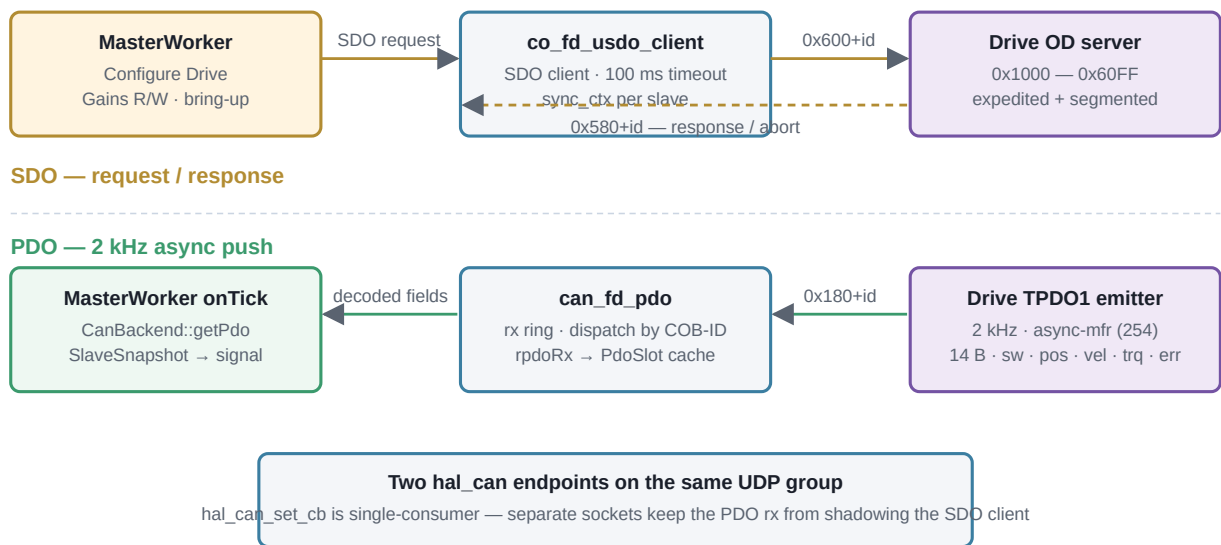
1. Drive Operation Enabled + correct mode.
2. Tuning dock → Step sub-tab → set params → **Run step**.
3. Read overshoot %, 10–90 % rise, SS error.
4. Iterate: bump Kp until overshoot, back off 20 %, add Ki.

Bode sweep

1. Tuning dock → Bode → freq start/end + duration.
2. **Run sweep** → chirp fires → Goertzel extracts I/Q per frequency.
3. Read −3 dB crossover + phase margin.

9. PDO mapping — read, edit, apply

Two wires per slave: SDO for commands, PDO for telemetry

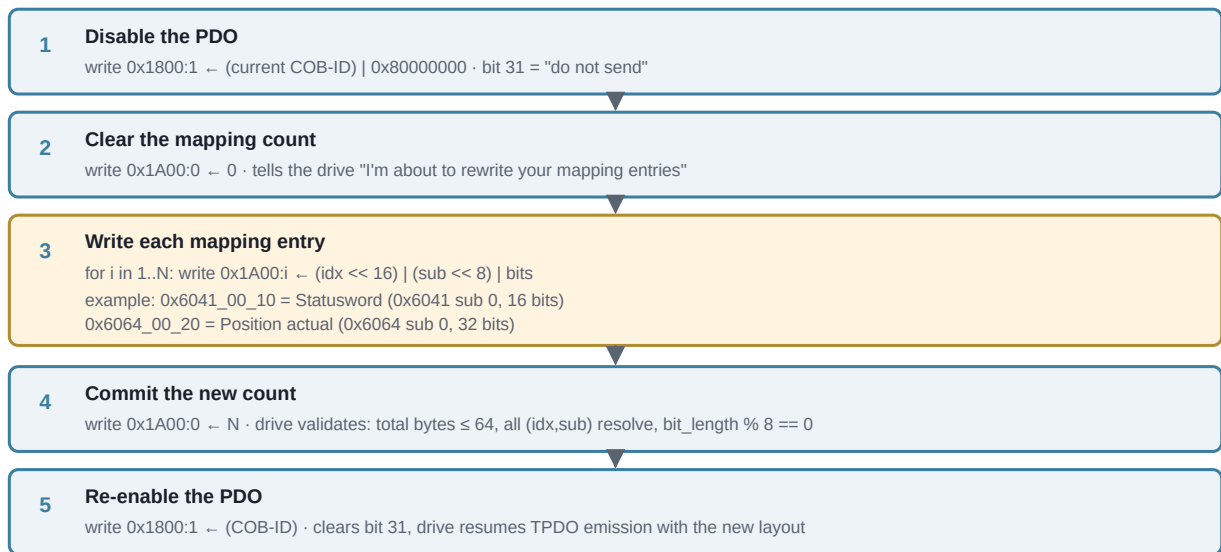


Two `hal_can` endpoints per connection — one SDO, one PDO — because `hal_can_set_cb` is single-consumer.

Edit flow: PDO map dock → **Edit...** → add/remove from CiA 402 catalog → OK. Under the hood:

CiA 301 §7.2.2.2 — PDO remap dance (TPDO1 example)

Every step is an SDO write from MasterWorker to the slave.



Against the simulator every step SDO-aborts (0x06020000) because the sim doesn't expose 0x1600 / 0x1A00 . Against real drives the sequence completes and the TPDO layout changes take effect on next emission.

10. Telemetry recording & export

- **Record telemetry** (Ctrl+Shift+R) — toggle; currently a placeholder (CSV-per-snapshot planned).
- **Export telemetry CSV...** — dumps buffered chart data.
- **Clear charts** — empty the scope.

11. Emergency stop

Red button, toolbar right edge. Space app-wide. On click: stop generator → master_mgr_disable_all → every slave → Switch On Disabled. Re-enable via normal Control tab Bringup + Enable.

12. Keyboard shortcuts

Combo	Action
Ctrl+K	Connect...
Ctrl+R	Reconnect
Ctrl+E	Edit motor profile
Ctrl+Shift+C	Configure Drive...
Ctrl+Shift+R	Toggle recording
Ctrl+O / Ctrl+S	Open / Save profile
Space	E-STOP
F1	Documentation

13. Troubleshooting

Symptom	Cause	Check
Configure Drive: "all fields aborted"	Drive doesn't expose these OD objects	Use a real drive
Session timeout	Drive not responding / wrong id / bus down	Verify id + transport
PDO Value column stays –	No TPDO — not emitting or COB-ID mismatch	Log "PDO online" line should appear
E-STOP disabled	Not connected	Connect first
Start Homing no effect	Drive not Operation Enabled	Enable on Control first
Floating dock off-screen	Window moved	Toggle from toolbar/View menu

Appendix A — Transports

UDP multicast (`hal_can_udp`)

Two sockets per connection; both bound `0.0.0.0:23400` with `SO_REUSEADDR`. Joined `239.192.0.42`, `IP_MULTICAST_LOOP=1`. 12-byte header `[tag·4][cob_id·4][dlc·1][flags·1][rsvd·2]`; tag filters self-echoes.

ZLG USB-CANFD (`hal_can_zlg`)

`libcontrolcanfd.so` via `dlopen()` at runtime; ships in `src/backends/zlg/lib/` (RPATH-wired). One endpoint per channel; PDO not yet plumbed (V1 is SDO-only).

Appendix B — CiA 402 OD quick reference

Index	Sub	Field	Where
0x1010	1	Store parameters	Save to Flash (planned)
0x1011	1	Restore defaults	Load from Flash (planned)
0x1400+n	1	RPDO COB-ID	PDO remap
0x1600+n	0..N	RPDO mapping	PDO mapping editor
0x1800+n	1	TPDO COB-ID	PDO remap
0x1A00+n	0..N	TPDO mapping	PDO mapping editor
0x603F	0	Error code	TPDO1
0x6040	0	Controlword	Control · Homing
0x6041	0	Statusword	TPDO1 · banners
0x6060	0	Mode of operation	Control · Homing
0x6064	0	Position actual	TPDO1 · Zero encoder
0x6065	0	Max following error	Protection
0x6066	0	Following error timeout	Protection
0x606C	0	Velocity actual	TPDO1
0x6072	0	Max torque	Protection
0x6077	0	Torque actual	TPDO1 · Zero torque
0x607A	0	Target position	RPDO1
0x607C	0	Home offset	Zero encoder here
0x607D	1,2	Position limits	Protection
0x6080	0	Max motor speed	Protection
0x6081	0	Profile velocity	Motion
0x6083	0	Profile acceleration	Motion
0x6084	0	Profile deceleration	Motion
0x6085	0	Quickstop deceleration	Motion
0x608F	1	Encoder resolution	Encoder
0x6091	1,2	Gear ratio	Encoder
0x6092	1,2	Feed constant	Encoder
0x6098	0	Homing method	Homing
0x6099	1,2	Homing speeds	Homing
0x609A	0	Homing acceleration	Homing
0x60B2	0	Torque offset	Zero torque here
0x60FF	0	Target velocity	RPDO1

Appendix C — Command-line flags

Flag	Value	Purpose
<code>-a</code> , <code>--auto-connect</code>	—	Connect with defaults on startup
<code>--quit-after</code>	ms	Quit after N ms (smoke tests)
<code>--screenshot</code>	path[:ms]	Grab window → PNG after delay
<code>--show-panel</code>	name	Open dock/dialog: <code>pdo-map</code> / <code>tuning</code> / <code>profile</code> / <code>drive-config</code>
<code>--show-tab</code>	idx	Select left-tab (0/1/2)
<code>--only-dialog</code>	—	With <code>--screenshot</code> : capture only the top dialog

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